tl;dr Notes on Nonlinear Wave Equations

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Fall 2026

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[0]Last updated: 2026-10-10 at 21:46

Overview

The following is a tl;dr version of the lecture notes on nonlinear wave equations I am writing in preparation for the SBU math PhD oral exam. The purpose of these notes aren't to include every single detail/computation that is inside the main set of notes, but rather to point out the important ideas, techniques, and perspectives.

1 Syllabus

Functional and Harmonic Analysis	
Harmonic Analysis	Three-Lines Lemma Riesz Interpolation Theorem Hausdorff-Young Inequality
	Extension of Fourier Transform (Bounded Map from $L^p \to L^q$ for all $p \in [1, 2]$)
	Young's Inequality for Convolutions Hilbert Transform

Nonlinear Wave Equations	
Linear Wave Equation	Cauchy Problem Well-Posedness Simple Energy Estimates Bootstrap Arguments Properties of Fundamental Solutions Explicit Representation Formulas
	Pointwise and Uniform Dispersion Rates ($n \geq 2$) (Faster Decay inside Bounded Ball vs. Globally)
	Finite Speed of Propagation Strong Huygens Principle
	Hahn-Banach Theorem and Existence of Weak Solutions (Variable Coefficient Equation)
Linear Hyperbolic Systems	Method of Characteristics and loss of uniqueness for unbounded speed
	Strictly/Symmetric Hyperbolic Systems
Nonlinear Wave Equations	**Local** Well-Posedness

2 Functional and Harmonic Analysis

- (**Three-Lines Lemma**) First interpolation result we obtain. For a holomorphic function Φ on the strip $S = \{0 < \mathrm{Re}(z) < 1\}$ and continuous on $\bar{S}$, and bounded by M_0 and M_1 on either side of the strip, Φ is also bounded on any vertical line between $\mathrm{Re}(z) = 0$ and $\mathrm{Re}(z) = 1$:

$$\sup_{y \in \mathbb{R}} |\Phi(t + iy)| \leq M_0^{1-t} M_1^t, \qquad \forall t \in [0, 1].$$

- (**Riesz-Thorin Interpolation Theorem**) Consider a linear operator T that is bounded from $L^{p_0} \to L^{q_0}$ and from L^{q_0} to L^{q_1}. Then this theorem gives us the conditions under which T is a bounded operator from L^p to L^q. In particular, one has that T is a bounded operator from L^p to L^q for all pairs (p, q) such that

$$\frac{1}{p} = \frac{1-t}{p_0} + \frac{t}{p_1}, \qquad \frac{1}{q} = \frac{1-t}{q_0} + \frac{t}{q_1}.$$

  Key Facts: (1) Reduces the effort to establishing endpoint estimates. (2) applications include the **Hausdorff-Young inequality**, **extension of the Fourier transforrm**, **Young's inequality for convolutions**.

3 Linear Wave Equation

- (**Cauchy Linear Wave Problem**) Let ϕ be a function (or distribution) of appropriate regularity on $I \times \mathbb{R}^n$, and let $\Box = \partial_{tt} - \Delta$ be the d'Alembertian operator. Then the Cauchy wave problem on $I \times \mathbb{R}^n$ is

$$\begin{cases} \Box \phi = f, \\ (\phi, \partial_t \phi) \restriction_{\{t=0\}} = (\phi_0, \phi_1) \end{cases},$$

  where f (potentially zero) is some source term.

- (**Well-Posedness of the Cauchy Problem**) A desired property for the Cauchy problem. We require the following three conditions:
  1. Existence of a solution;
  2. Uniqueness of the solution;
  3. Continuous dependence of the solution on the initial data (small perturbation in the initial data leads to a small perturbation in the solution).

- (**Energy of a Solution; Homogeneous**) Assuming a solution ϕ exists to $\Box\phi = 0$, define the energy $E(t)$ to be

$$E(t) = \frac{1}{2} \int_{\mathbb{R}^n} (\partial_t \phi)^2 + (\nabla \phi)^2 \, dx.$$

  (1) The energy is always conserved, and (2) gives uniqueness of solution (if a solution exists) by considering the difference of two solutions which must have zero energy at $t = 0$ (compare this to elliptic and parabolic equations, whose uniqueness of solutions can be given by a maximum principle).

- (**Energy Estimate of a Solution; Nonhomogeneous**) Assume $\partial_\alpha(a^{\alpha\beta}\partial_\beta\phi) = F$, where $a_{\alpha\beta}$ is some metric (close enough to the standard Minkowski metric $g_{\alpha\beta}$). Let $|\partial\phi|^2 \coloneqq (\partial_t\phi)^2 + (\nabla\phi)^2$. Although $\|\partial\phi\|_{C([0,T])}$ is not conserved, it is bounded above by the $\dot{H}^1(\mathbb{R}^n) \times L^2(\mathbb{R}^n)$ norm of the initial data, and the L^2 norm of the source term.

- (**Bootstrap Argument**) A method of establishing certain desired results. The idea is that if we wish to show that some result A is true, we start by assuming that some marginally better assumption A_ε is true, and showing that this better result forces the original result.

- (**Fundamental Solutions**) Distributional solutions to the wave equation,

$$\begin{cases} \Box E_+ = \delta_0, \\ \operatorname{supp}(E_+) \subseteq \{(t,x) : 0 \le |x| \le t\} \end{cases},$$

  I.e., a distribution E_+ solving $\Box E_+ = \delta_0$ that is supported on the forward light cone based at the origin. **Key Facts:** (1) the support of the fundamental solution grows with time (i.e., solutions to the wave equation propagation), (2) *if* a fundamental solution exists, then it propagates at finite speed, (3) a *unique* fundamental solution exists:

$$E_+(t,x) = -\frac{\pi^{\frac{1-n}{2}}}{2} \chi_{\{t\ge 0\}} \chi_+^{\frac{1-n}{2}} \left(t^2 - |x|^2\right),$$

  where

$$\chi_+^a(y) \coloneqq \frac{y^a \chi_{\{y\ge 0\}}}{\Gamma(a+1)};$$

  i.e.,

$$E_+(t,x) = -\frac{\pi^{\frac{1-n}{2}}}{2} \chi_{\{t\ge 0\}} \frac{1}{\Gamma(\frac{3-n}{2})} \frac{1}{(t^2-|x|^2)^{\frac{n-1}{2}}} \chi_{\{|x|\le t\}},$$

  (4) can be used to derive general representation formulas for the wave equation ($\Box\phi = F$):

$$\phi(t,x) = -E_+ * (\phi_1 \delta_{\{t=0\}}) - (\partial_t E_+) * (\phi_0 \delta_{\{t=0\}}) + (F \chi_{\{t\ge 0\}}) * E_+,$$

(5) can be used to understand pointwise decay estimates of solutions (pointwise, for every $R > 0$ and $|x| \leq R$, $|E_+(t,x)| \sim O(t^{1-n})$), (6) can be used to derive explicit representation formulas for solutions to the linear wave equation.

- (**Explicit Representation Formulas**) Solutions exist to linear wave equation, and are unique. Suffices for the initial data $(\phi_0, \phi_1) \in H^1(\mathbb{R}^n) \times L^2(\mathbb{R}^n)$.
- (**Decay Rates ($n \geq 2$)**) For homogeneous problem, conservation of energy and propagation of solution implies amplitude of wave decays in time. Locally (on a bounded ball), decay rate $\lesssim (1+t)^{1-n}$. Globally uniformly, decay rate $\lesssim (1+t)^{(1-n)/2}$. In particular, solution decays faster in a bounded ball than globally.